

Phase 8

Project Demonstration Phase

Phase 8: Project Demonstration Phase

The project demonstration video presents the complete working process of the AI WeatherWise backend application. The demonstration begins with an explanation of the project structure, major source files, technologies used, and the functionality of each backend component.

Code Explanation

The video explains the main implementation components, including:

- Node.js and Express.js – Used to create and manage the REST API server.**
- MongoDB and Mongoose – Used for database storage and schema management.**
- Authentication Middleware – Demonstrates JWT-based authentication and protected API access.**
- User and Favorite Location Models – Explains the database schemas and relationships used to manage user data and saved locations.**
- API Routes – Covers authentication, weather information, favorite location management, and AI-related routes.**
- Weather API Integration – Demonstrates how external weather data is retrieved and processed for users.**
- AI Integration – Demonstrates how AI is used to provide weather summaries and personalized weather recommendations.**
- Error Handling & Validation – Explains how invalid requests, authentication errors, and database/API errors are handled.**

API Testing

The second part of the demonstration focuses on testing the implemented APIs using Postman/Thunder Client. The following operations are demonstrated:

API Operation	Purpose
User Registration	Creates a new user account
User Login	Authenticates the user and generates a JWT token
User Profile	Retrieves the authenticated user's profile
Add Favorite Location	Saves a location to the user's favorite locations
Get Favorite Locations	Retrieves the user's saved locations
Current Weather	Retrieves current weather information for a selected city
AI Weather Summary	Generates an AI-based summary of the weather
AI Weather Recommendation	Provides AI-based weather recommendations

The API requests, request bodies, authentication tokens, and corresponding responses are shown during the demonstration. This confirms that the backend components are properly connected and the major functionalities of the AI WeatherWise application are working as expected. The project documentation also provides separate links for the API testing video and project demonstration.

Demo Video Link:

<https://drive.google.com/file/d/1b9DnvVokSZSs0F2gUH1QoKr0M1VxZxYd/view?usp=drivesdk>